

Education

University of Maryland, Baltimore County | Baltimore, MD | GPA:3.5

August 2023 – May 2025

Master of Science in Information Systems

Sinhgad Institute of Technology and Science| Pune, India | GPA: 3.5

August 2019 – June 2023

Bachelor of Engineering in Computer Science with Honors in Artificial Intelligence and Machine Learning

TECHNICAL SKILLS

- Programming language: Python, Java, C++, SQL, Oracle, MySQL
 - Machine Learning Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, PyTorch, Keras
 - Tools and Technologies: Power BI, Tableau, Seaborn, Weka, Hadoop, Jupyter Notebook
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WORK EXPERIENCE

Python Intern | Dataverse Infotech

September 2022 – February 2023

- Led Python application initiatives, increasing operational efficiency by 20% through coding standards and workflows.
- Augmented data processing by 30% using Pandas and NumPy for high-performance data manipulation and analysis.
- Conducted rigorous code reviews, enhancing system performance by 23% through refined algorithms and optimizations.
- Improved technical operations by 18% via continuous code optimization, system upgrades, and workflow enhancements.

Data Science Intern | LetsGrowMore

December 2021 – January 2022

- Built predictive models using regression, boosting forecast accuracy by 15% and providing insights for strategic planning.
 - Developed a market trend forecasting system with TensorFlow and Scikit-Learn, improving prediction accuracy by 20%.
 - Created interactive Tableau dashboards, enabling stakeholders with accessible insights and enhancing decisions by 25%.
 - Implemented segmentation strategies to boost customer retention, achieving a 28% increase in engagement and retention.
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PROJECTS

Sign Language and Emotion Detection

- Formulated a real-time Emotion Detection model using Convolutional Neural Networks (CNNs) with TensorFlow and Keras, achieving 92.5% accuracy in facial expression classification across diverse datasets.
- Enhanced and augmented 50,000+ facial images using OpenCV and NumPy, applying data augmentation techniques (rotation, flipping, scaling) to improve model generalization by 18% for real-world scenarios.
- Integrated model deployment with Flask and TensorRT, reducing inference time by 40% and delivering real-time emotion recognition with response times of less than 150ms for seamless user experiences.
- Engineered an interactive visualization dashboard using Matplotlib and Seaborn, offering comprehensive model performance insights and detailed facial emotion analytics to enhance result explainability.

Crime Analysis | [GitHub](#)

- Developed a crime prediction model using Machine Learning (Random Forest, XGBoost, Decision Trees), achieving 87.2% accuracy and enhancing predictive analytics capabilities for law enforcement operations.
- Analyzed over 6 million crime records using SQL, Pandas, and NumPy, applying feature engineering, geospatial clustering (K-Means, DBSCAN), and time-series forecasting (LSTM, ARIMA) to identify high-risk zones.
- Designed interactive dashboards with Tableau and Power BI, integrating Google Maps API and Folium to visualize 200+ crime hotspots, reducing analysis time by 40% and enhancing real-time decision-making processes.
- Deployed a Flask-based web application with real-time crime insights, optimizing database queries (PostgreSQL, MySQL) and achieving a response time of under 200ms, significantly improving public safety awareness.

Traffic Flow Optimization Project

- Designed an AI-powered Smart Traffic Light Management System using Computer Vision and Deep Learning (OpenCV, TensorFlow), significantly reducing traffic congestion by 35% through adaptive signal control.
 - Processed and analyzed 100,000+ real-time traffic images using NumPy and Pandas, applying advanced image preprocessing techniques to improve object detection accuracy by 22%.
 - Implemented and deployed an adaptive traffic control algorithm integrating real-time vehicle detection, resulting in a 40% reduction in average wait times at intersections across multiple traffic zones.
 - Constructed an interactive dashboard using Power BI and Matplotlib, visualizing complex traffic density patterns and optimizing signal timings with live data feeds, improving traffic flow efficiency by 30%.
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MACHINE LEARNING PUBLICATIONS

[Medication system for pharma store](#) | 2023 JETIR June 2023, Volume 10, Issue 6

Journal of Emerging Technologies and Innovative Research (JETIR)

ISSN No: - 2349-5162

[Survey on Classification Techniques in Healthcare](#) | Volume 8, Issue 3, March – 2023

International Journal of Innovative Science and Research Technology

ISSN No: -2456-2165

CERTIFICATES & COURSES

SQL(Intermediate) | HackerRank (February 2025)

IBM Skills Build Innovation Camp | IBM (July 2021)

Software Engineer| HackerRank (February 2025)

Foundations: Data, Data, Everywhere | Google (November 2021)
